Runge-Kutta Low Storage Methods

for Spacecraft Thermal Simulation

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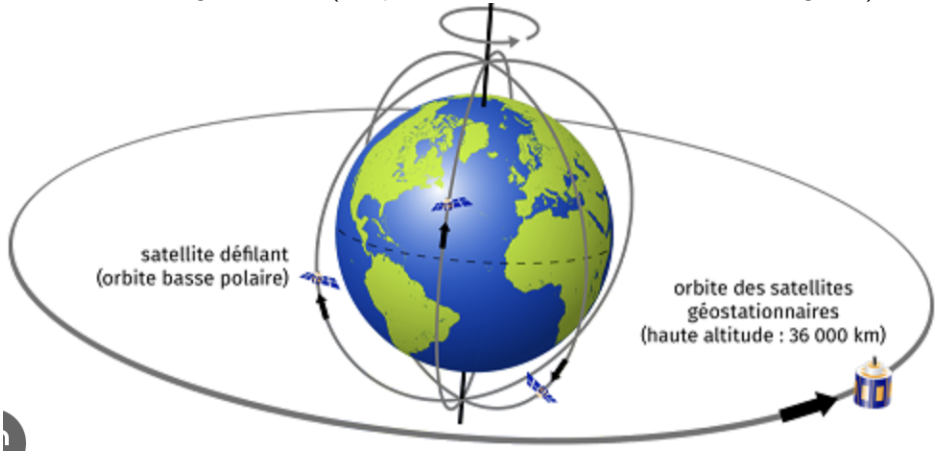
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Analysis, Implementation and Benchmarking of Advanced Numerical Methods

Context

Satellites orbiting the Earth (temperatures from -150 to 150 Celsius degrees.)



Context

Temperature equipement margins (-20 to 50 Celsius degrees) .



Industrial Context — DOREA (SME) & Thermal Simulation

DOREA (Sophia Antipolis): Spacecraft thermal solver development.

Orbital Environment & Challenges:

- Direct solar flux (Low Earth Orbit).
- Rapid alternation between sunlit and eclipse phases.
- Large coupled models: 10^4 to 10^5 thermal nodes per satellite.

Limitations of Classical RK4 Solver:

- Memory overhead: Requires storing 4 intermediate vectors ($4n$ floats).
- Performance degrades due to memory bottlenecks on large coupled systems.

Outline

Numerical Methods

Parallelism & Amdahl's Law

Benchmarking

Spacecraft Application

Conclusion

Spacecraft Thermal Equation — Nodal Method

Energy balance for node i:

$$M_i C_i \frac{dT_i}{dt} = \underbrace{\alpha_i S_i \phi_S}_{\text{Solar}} + \underbrace{\alpha_i S_i \phi_\alpha}_{\text{Albedo}} + \underbrace{\varepsilon_i S_i \phi_T}_{\text{Earth IR}} + \underbrace{Q_i}_{\text{Dissip.}} + \underbrace{\sum_j GL_{ij}(T_j - T_i)}_{\text{Conduction}} + \underbrace{\sum_j \sigma\, GR_{ij}(T_j^4 - T_i^4)}_{\text{Radiation}}$$

Key characteristics:

- **External fluxes**: Time-dependent (sunlit vs. eclipse phases).
- **Conduction**: Linear, drives potential stiffness based on material capacity.
- **Radiation**: Nonlinear (T^4), necessitating an advanced solver like RK4.

Numerical Methods: Euler, RK2, RK4 Classic

Cauchy problem:

$$\frac{d\mathbf{y}}{dt} = f(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0$$

Euler (Order 1)

- Global error: $\mathcal{O}(h)$
- 1 evaluation / step
- 0 temporary vectors

RK2 - Heun (Order 2)

- Global error: $\mathcal{O}(h^2)$
- 2 evaluations / step
- 2 temporary vectors

RK4 Classic (Order 4)

- **Currently used by DOREA**
- Global error: $\mathcal{O}(h^4)$
- 4 evaluations / step
- **4 temporary vectors** ($4n$ overhead)

RK4 Low Storage (Carpenter & Kennedy, 1994)

The 5-stage recurrence ($s = 1, \ldots, 5$):

$$\mathbf{d} \leftarrow \alpha_s \, \mathbf{d} + h \, f(t_n + c_s h, \, \mathbf{y}_i)$$

$$\mathbf{y}_i \leftarrow \mathbf{y}_i + \beta_s \, \mathbf{d}$$

Final step:

$$\mathbf{y}_{n+1} = \mathbf{y}_i$$

Algorithmic Advantages (The 2N-Method):

- **Order 4 preserved**: Rigorously 4th-order accurate.
- **Reduced Memory**: Only 2 working vectors ($\mathbf{d}$ and $\mathbf{y}_i$) instead of 4.
- Memory overhead drops from $4n$ to $2n$ ($\approx 50\%$ reduction in working storage).
- M. H. Carpenter and C. A. Kennedy, Fourth-Order 2N-Storage Runge-Kutta Schemes, NASA Technical Memorandum 109112, NASA Langley Research Center, 1994

Parallel RK4-LS & Amdahl's Law

Parallelisation Strategy:

- Multi-threaded evaluation of vector operations via Python's `numexpr`.
- Data parallelism: n components distributed across CPU threads.
- **Limitation:** The time loop remains fundamentally sequential.

Amdahl's Law:

$$S(p) = \frac{1}{1 - p_{\text{par}} + \frac{p_{\text{par}}}{p}}$$

As $p \to \infty$, the speedup is bounded by the irreducible sequential fraction $(1 - p_{\text{par}})$.

Benchmark: Accuracy, Memory, and Time

Nonlinear test system:

$$f(t, \mathbf{y}) = 0.5 \sin(\mathbf{y}_{\text{roll}}) - 0.7 \cos(\mathbf{y}) + 0.1 \sin(t/20)$$

Accuracy ($n = 10,000$):

- Euler: $\approx 1.68 \times 10^{-5}$
- RK2: $\approx 7.40 \times 10^{-7}$
- RK4 Classic: $\approx 8.80 \times 10^{-9}$
- **RK4-LS**: $\approx 8.99 \times 10^{-9}$

RK4-LS achieves identical 4th-order accuracy.

Memory ($n = 500,000$):

- RK4 Classic: 27,345 KB
- **RK4-LS**: 19,532 KB (**-28% overhead**)

Computation Time (Python):

- RK4-LS is $\approx 35\%$ slower than RK4 Classic due to 5 stages vs 4. Gap narrows in compiled languages (Fortran).

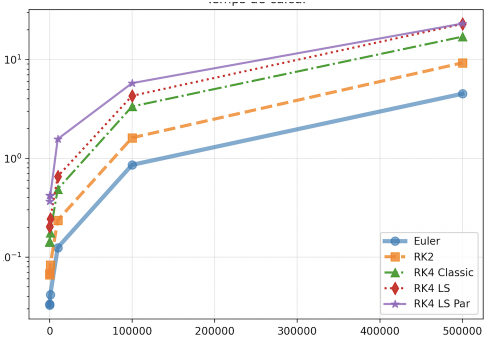
Benchmark: Parallelism Validation

Parallel vs Serial computation time (RK4-LS):

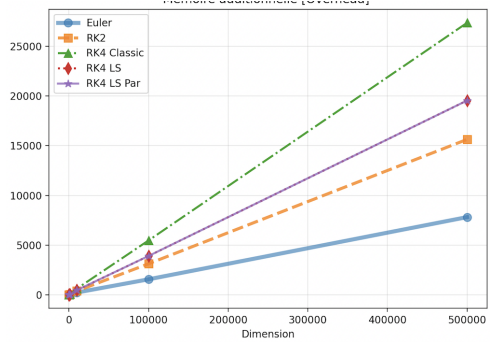
Dimension (n)	Serial (s)	Parallel (s)	Ratio (Par/Ser)
10	0.20	0.42	$\times 2.07$
1,000	0.24	0.42	$\times 1.71$
10,000	0.66	1.57	$\times 2.39$
100,000	4.30	5.78	$\times 1.34$
500,000	**22.99**	**23.18**	$\times 1.01$

- `numexpr` thread overhead exceeds computational benefit for $n \leq 100,000$.
- Break-even point is beyond 10^6 components.
- Confirms Amdahl's Law: sequential time loop caps the maximum speedup.

Performance Benchmark Results: Time & Memory

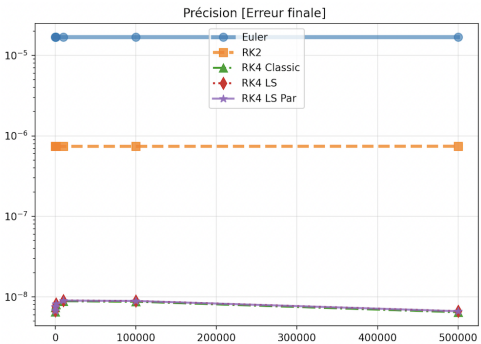


Computation Time:

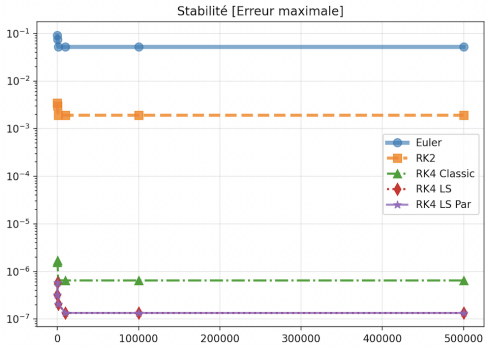


Additional Memory:

Performance Benchmark Results: Accuracy & Stability



Final Error:



Stability: Maximum error over time

Matrix ODE Formulation & Simulation

Compact matrix form for the entire spacecraft:

$$\mathbf{MC}\,\frac{d\mathbf{T}}{dt} = \mathbf{P}(t) + C_{\mathrm{cond}}\mathbf{T} + \sigma R_{\mathrm{rad}}\mathbf{T}^4$$

LEO Orbit Simulation Parameters:

- $t_f = 5,400$ s (1 LEO orbit, ≈ 90 min).
- Alternation: 60% sunlit phase, 40% eclipse phase.
- Time step set to $h = 0.5\,h_{\mathrm{CFL}}$ to guarantee stability.

Observed Physical Behaviour:

- Temperatures rise during sunlit phase and cool via radiation during eclipse.
- RK4-LS remains perfectly stable throughout the nonlinear radiative regime.

Conclusion & Perspectives

Key Takeaways:

1. **Accuracy:** RK4-LS achieves the same 4th-order precision as Classical RK4.
2. **Memory:** Working memory overhead is reduced by 28% (from 4 down to 2 intermediate vectors), crucial for heritage Fortran solvers.
3. **Parallelism:** Validated theoretical limits; multi-threading vector operations is only viable beyond $\approx 500,000$ components due to Amdahl's Law.
4. **Application:** The method correctly and stably handles the nonlinear spacecraft nodal matrix system.

Perspectives:

- Explore higher-order LSRK schemes (order 5 or 6).
- Implement GPU-based parallelisation for extremely large thermal models.

Thank you for your attention

Questions?

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